TRANSFINITTE · HARDWARE TRACK · FINAL DESIGN

HridayMitra: a heart check-in that fits an Indian home

A seated 2-minute daily check-in station, a chest pendant worn like a locket, and a pill tray, all feeding one alert engine that warns the family and the ASHA worker days before a hospital emergency.

Monitoring only, not a diagnosis tool. It does not replace a doctor or emergency care.

1. The idea in one minute

Most teams will pick one sensor: a scale, a wristband, a bed sensor or a phone app. HridayMitra covers each sensor's weak spot with another part, so the patient never has to do much and no single failure breaks the system.

Part	What it is	Patient effort	What it measures
1. Check-in station	A stool standing on a weighing platform, with 4 steel foot plates, a handle with a finger clip, a speaker, 3 lights and 2 big buttons	About 90 s each morning and 20 s at evening pill time, seated	Weight; leg-fluid impedance at 3 frequencies; pulse and SpO_2; 2 yes/no questions
2. Chest puck (on a cord from the station)	Palm-sized puck with 4 steel electrodes and a contact microphone	Press to the chest for 20 s with the morning check	**Chest impedance (lung fluid)**, lung sounds and the S3 heart sound
3. Chest pendant	A locket on a cord: ESP32-C3, motion sensor, small battery, SOS button	Wear it, nothing else	Night breathing rate and pattern, **body angle (lying flat, propped up, sitting up)**, activity, falls, SOS
4. Pill tray (built into the station)	Weekly tray with a sensor per slot and a spoken reminder	Take pills as usual	Which doses were taken or missed

The station runs the alert logic itself. It does not need internet. Alerts go out by **SMS and an automatic voice call** in the local language. The pendant stores 7 days of data and hands it to the station when they are in range, so power cuts and trips away do not lose data.

2. A day with HridayMitra

When	What happens	Time
Morning, after the toilet	Patient sits on the stool. Voice: "Namaste Ramesh-ji, please take off your slippers." Asks: "Did you get breathless when you bent down? Green for no, red for yes." Feet on the plates, finger in the clip, hold the handle (about 40 s). Press the puck to the chest (about 20 s). Asks: "Did you need more pillows or sit up to breathe last night?" Shows a green, amber or red light and a short spoken result. Then reminds the patient to take the morning pills (the tray never locks).	~90 s
Evening pill time	A 20-second foot-only reading. The drop in leg fluid from morning to evening is compared with the patient's own normal. Asks: "Did you take your pills?"	~20 s
Day and night	The pendant records quietly. SOS button and fall detection work at home. Away from home it only logs.	0 s

Total: under 2 minutes a day. Family members can step on the station and are logged as guests, never counted for the patient.

3. How every PS warning sign is measured

PS warning sign	Main measurement	Backup measurement
Weight gain	Station load cells (seated, stool and handle on the platform)	Leg impedance trend
Faster resting heart rate	Finger sensor at the station, seated at rest	Pendant chest vibration at night (stretch goal, untested)
Faster breathing	Pendant, at night (rate and rapid-shallow pattern)	Station breathing estimate (weak, used as a multi-day trend)
Less activity	Pendant activity and step minutes	Missed check-ins, missed pills
Fluid in legs	**Foot impedance at 3 frequencies (about 5, 50 and 250 kHz).** The ratio of low to high frequency shows extracellular fluid, so skin contact and body size largely cancel out.	**Morning versus evening** change in the same patient (cancels contact and skin differences); weight; "feet more swollen?" question
Fluid in lungs	**Chest-puck impedance** (falls as lung fluid rises)	Night breathing pattern, lung sounds and S3, lying-flat signals, SpO_2, symptoms (the Lung Congestion Score, below)
Trouble breathing lying flat	**Pendant body angle:** nights spent propped up or sitting up, compared with the patient's own normal	"More pillows than usual?" question

Lung Congestion Score. No ₹5,000 device can measure lung water directly. So the score fuses five independent signals, each compared with the patient's own normal: (1) chest impedance, (2) night breathing rate and pattern, (3) lung sounds and S3, (4) lying-flat angle plus the pillow question, (5) SpO_2 and symptoms. A lung alert needs at least two of them to move together plus one non-lung sign. Each alert says which signals moved.

Evidence, honestly: a chest-impedance device identified radiographic lung fluid in emergency patients with 90% sensitivity and 94% specificity (one older ED study); implanted impedance managed 60% and 73%. A scale adding foot impedance caught 70% of events against 35% for weight alone (SCALE-HF 1, company-linked). Neither has been tested with our hardware. Real accuracy at home is unknown.

4. Alert engine and who gets told

- **Personal baseline:** rolling median and spread of each signal from days 7 to 35 back. Only worsening counts (weight up, impedance down, heart rate up, breathing up, SpO_2 down, activity down), so one improving signal cannot cancel another. A new patient starts from the hospital's "dry weight" and population averages, so the first weeks are covered.
- **Three signal families:** fluid (weight, leg and chest impedance), cardio-respiratory (heart rate, breathing, SpO_2) and symptoms and behaviour (questions, missed check-ins, missed pills, activity).
- **Amber** = one day above the threshold. The device asks for a repeat check that evening or next morning.
- **Red** = above threshold on 2 of the last 3 days *and* at least 2 families contributing. It clears only when well below the threshold, so alerts do not flicker (hysteresis, as in HeartLogic).
- **Immediate danger rules** (confirmed by one quick repeat reading): SpO_2 below 90%, "breathless at rest", very fast or slow pulse, or SOS.
- **Cause tags:** HF-like, infection-like (heart rate and breathing jump, weight flat), salty-meal-like (weight up then back), after missed medicine, dehydration-like. **A missed water pill never silences an alert.** It changes the advice: "Medicine missed 2 days. Please take it and call ASHA."
- **Irregular pulse** from the finger sensor is flagged as "please ask a doctor", not counted as an HF alert, because pulse-only rhythm checks give many false positives in older people.

Level	Patient (device)	Family	ASHA worker	Doctor
Green	Green light, "All fine"	Weekly "device OK" SMS	—	Monthly summary (urban: dashboard)
Missed check-in	Reminder next time	SMS after about 11 am; voice call after 2 days	SMS after 2 days	—
Amber	Amber, "check again this evening"	SMS	—	—
Red	Red, spoken reason	Automatic voice call in the local language plus SMS with reason	SMS with patient name and reason	SMS summary; urban: dashboard and eSanjeevani link
Urgent	Flashing red, "Go to hospital, call 108"	Repeated calls to 2 contacts	SMS	SMS

Nothing ever calls an ambulance automatically. A person confirms. Silence is never shown as "all fine": a missed check-in, a dead battery or a forgotten pendant always produces a message.

5. Edge cases and false alerts

Problem	Answer in the design
Another person uses the station	Weight and impedance must fit the patient's recent range. If not, it asks "Is this Ramesh-ji?" or logs a guest that never enters the baseline. The pendant and pill tray are personal.
Blankets, extra pillow, shared bed	The station is used seated and barefoot, so bedding does not matter. The pendant compares each night with the patient's own normal, and the question asks about a *change* in pillows.
Sudden problem during the day	SOS and fall detection on the pendant and station, plus immediate danger rules. Away from home the pendant only logs.
Forgot the pendant or a check-in	The station still works alone, and the missing data is flagged. A missed check-in triggers a reminder, then an SMS.
Wet feet, slippers, dry heels	4-electrode measurement cancels most contact error. A contact check asks the patient to dry the feet or remove slippers.
Puck not placed properly	Contact check by voice, up to 2 retries. If it fails, the lung score uses the remaining signals and notes the gap.

Salty meal, infection, medicine change	Two-of-three-days rule, cause tags, a family SMS keyword to mark a medicine change.
Power cut, weak internet	Battery-first design, on-device logic, SMS and voice calls without data, readings stored and sent later.
Cannot read, any language	Voice prompts in the local language, 3 lights, 2 buttons. Nothing to read.
Rural and urban homes	Same device. Urban homes can add Wi-Fi sync and a web dashboard for children living elsewhere.
Device fault	Self-test each check-in (reference resistor, weight zero, battery), plus a weekly "device OK" SMS.

6. Cost (planning estimates, in ₹)

Item	₹	Status
ESP32 classic DevKit (has the signal generator)	600	Not verified
4 load cells and HX711	700	Verified range 544–880
Platform, stool, enclosure	450	Estimate
4 steel foot plates	150	Estimate
Multi-frequency impedance circuit (DDS chip, op-amps, switch, reference resistors)	500	Estimate
Finger pulse and SpO_2 sensor (MAX30102)	150	Weak listing at 96
Chest puck (4 electrodes, contact mic, cable, case)	350	Estimate
Speaker and amplifier	320	Amp 249 verified
Buttons and LEDs	150	Estimate
One 18650 battery, charger, boost converter	300	Estimate
Wires and PCB	200	Estimate
2G modem (SIM800L) for SMS and calls	300	150–448 verified
Pill tray sensors and box	150	Estimate
Chest pendant (ESP32-C3, motion sensor, small battery, SOS button, case)	800	Estimate
Total, planning prices	**5,120**	**Over ₹5,000**
With a hacked cheap bathroom scale replacing the load cells and platform (saves about ₹500)	**~4,620**	**Under, if the saving is real**

Go / no-go rule for P1 in the first two days: find a real bathroom scale that cuts at least ₹500. If it does not, **drop the pendant** (about ₹4,320) and move night breathing and lying-flat questions to symptoms only. The ₹5,000 limit is a hard rule.

The SIM800L does not work on Jio SIMs. Demo with an Airtel, Vi or BSNL SIM. A 4G modem (about +₹600–800) is the product version and needs a custom board to stay near budget. SIM and recharge are a running cost, not a part.

7. How it meets the PS and the judging

PS requirement or criterion	How
Under 2 minutes a day, no all-day patch	About 110 s seated; the pendant is worn, not stuck on
Little reading, local language	Voice, 3 lights, 2 big buttons
Power cuts, weak internet	Batteries, offline engine, SMS and voice calls
Parts under ₹5,000	About ₹4,620 planning with the go/no-go rule
Sensing method, hardware, software	Parts 1–4 above and the alert engine
Personal baseline, explained alerts, simple screen	Section 4; lights and voice for patients; SMS and calls for family; small web dashboard for doctors
Block diagram and schematic with sensor and microcontroller	Custom part: the multi-frequency impedance front end with the ESP32 (shared by the foot and chest channels)
Simulation proof	LTspice: current source and body model with a contact-resistance sweep (wet and dry feet) and a fluid-change sweep. Wokwi: check-in firmware and pendant firmware. Python (NeuroKit2/SciPy): signal pipeline.
Validation	Section 8
Public GitHub, 2-page report, 5-minute pitch	Owned by P5 with the others contributing
Creativity of sensing (20)	Seated closed-loop station; three-frequency foot ratio with morning/evening change; chest puck; passive lying-flat detection; pill-tray adherence loop
Ease and long-term use (20)	Habit-anchored: the pill reminder is the reason to use it; voice-guided; guest-friendly; silence never counts as normal
Software (20)	Fusion engine with a Lung Congestion Score, cause tags and ablations
Hardware and simulation (15)	Custom impedance circuit with simulation proof
Science and honest testing (15)	Cited parameters, patient-level split, baselines, stated limits
Cost, India fit, pitch (10)	Itemised ₹ list; ASHA and eSanjeevani path; 2G/4G choice

8. Validation plan

- **Data:** at least 200 synthetic patients, each with a 30-day warm-up plus a 30-day test window (the PS minimum is 20). Worsening starts 3–10 days before the hospital visit, so median lead time cannot exceed 10 days. 30–40% of events show no weight gain. Model: infection, atrial fibrillation, salty meals, medicine changes, missed pills, guest users, wet feet and missing days. Every size of change is written down in `DATA.md`, with its source or marked "assumption".
- **Real data:** test the heart-rate code on the BIDMC CHF database.
- **Split:** by patient (70% tuning, 30% test, run once). Report 95% confidence intervals.
- **Report all four PS numbers:** sensitivity, specificity, median lead time, false alerts per patient per month, plus time in alert and repeat checks requested.
- **Baselines to beat:** weight rules (about 23–35% sensitivity, about 0.35 false alerts per patient-month). Target: higher sensitivity at no more than 0.21 false alerts per patient-month. Run ablations: weight only, then add impedance, chest, breathing, symptoms, pills.
- **Honesty rule:** synthetic results only show the engine finds the patterns we built in. They do not prove clinical accuracy. Say so on the results slide. A Phase 2 pilot would compare our lung score with a doctor's lung-ultrasound B-line count.

9. Who builds what (5 people, in parallel)

Person	Owns
P1 Hardware	Block diagram; schematics of the impedance front end, puck and pendant; LTspice and Wokwi proofs; parts list with links; the bathroom-scale go/no-go
P2 Signals	Finger pulse and SpO_2 with quality checks; breathing rate, periodic breathing and body angle from the pendant motion data; chest-sound features; BIDMC heart-rate test
P3 Synthetic data	200+ patients with leg ratio (morning and evening), chest impedance, night breathing, angle, adherence and all confounders
P4 Alert engine	Baselines, scores, two-step confirmation, hysteresis, cause tags, SMS text, evaluation against weight rules
P5 Product	Voice prompts in local languages, family and doctor dashboard, user journey, 2-page report, pitch, GitHub

Fixed file formats between roles (daily features, answer key, alerts) were set in the earlier team page; the daily-features columns need updating for the new signals.

10. Limits and risks, stated plainly

- **No claim of 100% accuracy.** Even implanted devices catch about 70% of events. Fluid in the lungs is estimated, not measured directly.
- **Unproven parts:** our own impedance circuit, chest-puck placement by elderly users, pendant breathing and angle quality, and the pendant's battery life (my estimate: weeks). Pulse-sensor quality on cold, dry or shaking fingers is also unknown.
- **Cost:** a third of the parts list is estimated. The limit only holds if the scale saving is real.
- **Safety:** impedance current for people with pacemakers or defibrillators was not researched. The design includes an "impedance off" setting for the doctor to use.
- **It cannot predict sudden death** or replace emergency care. The response loop (family, ASHA, doctor) decides outcomes more than the sensors do.
- **Regulation:** a device claiming early warning would likely need CDSCO registration (probably Class B or C). Start by selling it as a scale with reminders and family notification.

Key sources

SCALE-HF 1 foot-impedance scale · Thoracic impedance vs chest X-ray · Zink 2020, segmental impedance in heart failure · HeartLogic (MultiSENSE) · Gyllensten 2016 trend detectors · Chaudhry 2007, weight before admission · Lung ultrasound in heart failure · Wearable lung-sound and fluid patch · Cordio voice study (stretch goal) · Kerala heart failure registry · Medicine non-adherence and decompensation

Several numbers come from abstracts, trade press or company-linked studies. Check the full papers before quoting them in the pitch.